

A design perspective of instrumental variable method

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Instrumental variable method

- A general approach to identifying causal effects with latent confounding
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- Two perspectives
 - design perspective: encouragement design
 - econometric perspective: two-stage least squares

Encouragement Design

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 - ① some in the treatment group refuse to take the treatment
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- **Encouragement design**: randomize the encouragement to receive the treatment rather than the receipt of the treatment itself
- Can we estimate the treatment effect (effect of the actually received treatment)?

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- Potential outcome: $Y_i(z)$
- Observed treatment receipt indicator: $D_i = D_i(Z_i)$
- Observed and potential outcomes: $Y_i = Y_i(Z_i)$

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 - comparison of the treated and untreated subjects who follow the treatment assignment
- What's the definition of treatment effect?

Compliance behavior

- Four principal strata (latent types):
 - compliers $(D_i(1), D_i(0)) = (1, 0)$,
 - non-compliers $\begin{cases} \text{always-takers} & (D_i(1), D_i(0)) = (1, 1), \\ \text{never-takers} & (D_i(1), D_i(0)) = (0, 0), \\ \text{defiers} & (D_i(1), D_i(0)) = (0, 1) \end{cases}$
 - denote the compliance behavior (a, n, c, d) by $U_i \rightsquigarrow D_i$ is a function of Z_i and U_i

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 - denote the compliance behavior (a, n, c, d) by $U_i \rightsquigarrow D_i$ is a function of Z_i and U_i
- Observed strata and compliance behavior:

	$Z_i = 1$	$Z_i = 0$
$D_i = 1$	Complier/Always-taker	Defier/Always-taker
$D_i = 0$	Defier/Never-taker	Complier/Never-taker

Assumptions

- **Randomization:** $\{Y_i(1), Y_i(0), D_i(1), D_i(0)\} \perp\!\!\!\perp Z_i$
- **Monotonicity:** $D_i(1) \geq D_i(0) \rightsquigarrow$ no defiers
- **Exclusion restriction:** $Y_i(1) = Y_i(0)$ for always takers and never-takers
- $\mathbb{E}\{D_i(1) - D_i(0)\} > 0 \rightsquigarrow$ there exists compliers

Monotonicity: implications

- Observed strata and compliance behavior under monotonicity

	$Z_i = 1$	$Z_i = 0$
$D_i = 1$	Complier/Always-taker	Always-taker
$D_i = 0$	Never-taker	Complier/Never-taker

- $\text{pr}(\text{never-taker}) = \text{pr}(D_i = 0 \mid Z_i = 1)$
 $\text{pr}(\text{always-taker}) = \text{pr}(D_i = 1 \mid Z_i = 0)$
 $\text{pr}(\text{complier}) = \text{pr}(D_i = 1 \mid Z_i = 1) - \text{pr}(D_i = 1 \mid Z_i = 0)$

Exclusion restriction: implications

- ITT effect for units with different compliance behavior

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 - always-taker: $ACE_a = \mathbb{E}\{Y_i(1) - Y_i(0) \mid D_i(1) = D_i(0) = 1\} = 0$
 - never-taker: $ACE_n = \mathbb{E}\{Y_i(1) - Y_i(0) \mid D_i(1) = D_i(0) = 0\} = 0$
 - complier: $ACE_c = \mathbb{E}\{Y_i(1) - Y_i(0) \mid D_i(1) = 1, D_i(0) = 0\}$

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- ITT effect for units with different compliance behavior
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 - complier: $ACE_c = \mathbb{E}\{Y_i(1) - Y_i(0) \mid D_i(1) = 1, D_i(0) = 0\}$
- ACE_c : **Complier Average Causal Effect (CACE)** or **Local Average Treatment Effect (LATE)**
 - for compliers: $Z_i = D_i$, ITT effect = treatment effect

Complier average causal effect

- ITT effect decomposition:

$$\begin{aligned}ACE_Y &= ACE_c \times \text{pr}(\text{compliers}) + ACE_a \times \text{pr}(\text{always-takers}) \\ &\quad + ACE_n \times \text{pr}(\text{never-takers}) \\ &= ACE_c \times \text{pr}(\text{compliers})\end{aligned}$$

- ITT effect on D : $ACE_D = \text{pr}(\text{compliers})$

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- ITT effect on D : $ACE_D = \text{pr}(\text{compliers})$

- $ACE_a = ACE_Y / ACE_D$

Identification

- ACE_Y and ACE_D are identifiable
- Identification formula:

$$\begin{aligned} ACE_c &= \frac{\mathbb{E}(Y_i \mid Z_i = 1) - \mathbb{E}(Y_i \mid Z_i = 0)}{\mathbb{E}(D_i \mid Z_i = 1) - \mathbb{E}(D_i \mid Z_i = 0)} \\ &= \frac{\text{Cov}(Y_i, Z_i)}{\text{Cov}(D_i, Z_i)} \end{aligned}$$

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- Z acts as an IV for D

Complier average causal effect

- $CACE = \mathbb{E}\{Y_i(1) - Y_i(0) \mid D_i(1) = 1, D_i(0) = 0\}$
 - average effect of encouragement for compliers
 - average treatment effect for compliers ($Z_i = D_i$)

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- $CACE \neq$ treatment effect unless the treatment effect for non-compliers equals CACE
- Different encouragement yields different compliers

Alternative notation

- Potential outcome: $Y_i(z, d)$
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- Exclusion restriction: $Y_i(z, d) = Y_i(d)$
 - $ACE_Y = \mathbb{E}\{Y_i(D_i(1)) - Y_i(D_i(0))\}$
 - $ACE_c = \mathbb{E}\{Y_i(d = 1) - Y_i(d = 0) \mid U_i = c\}$

Inference

- Wald estimator: $\widehat{CACE}_{\text{Wald}} = \frac{\widehat{ACE_Y}}{\widehat{ACE_D}}$

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Inference

- **Wald estimator:** $\widehat{CACE}_{\text{Wald}} = \frac{\widehat{ACE}_Y}{\widehat{ACE}_D}$
- Consistency: $\widehat{CACE}_{\text{Wald}} \xrightarrow{p} CACE = ACE_c$
- Asymptotic variance via the Delta method:

$$\text{var}(\widehat{CACE}_{\text{Wald}}) \approx \frac{1}{\widehat{ACE}_D^4} \left\{ \widehat{ACE}_D^2 \text{var}(\widehat{ACE}_Y) + \widehat{ACE}_Y^2 \text{var}(\widehat{ACE}_D) - 2 \widehat{ACE}_Y \widehat{ACE}_D \text{cov}(\widehat{ACE}_Y, \widehat{ACE}_D) \right\}.$$

where

$$\text{cov}(\widehat{ACE}_Y, \widehat{ACE}_D) = \frac{\text{cov}(Y_i(1), D_i(1))}{n_1} + \frac{\text{Cov}(Y_i(0), D_i(0))}{n_0}$$

Variance calculation

$$\widehat{\text{CACE}}_{\text{Wald}} - \text{CACE} \approx \frac{\widehat{\text{ACE}}_Y - \text{CACE} \cdot \widehat{\text{ACE}}_D}{\text{ACE}_D}$$

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- $\widehat{ACE}_Y - CACE \cdot \widehat{ACE}_D$ is the difference in means estimator for the adjusted outcome $A_i = Y_i - CACE \cdot D_i$
- Variance estimation steps
 - obtain the adjusted outcome $A_i = Y_i - \widehat{CACE} \cdot D_i$
 - obtain the variance estimate based on the adjusted outcome

$$\widehat{V}_A = \frac{\widehat{\text{var}}\{A_i(1)\}}{n_1} + \frac{\widehat{\text{var}}\{A_i(0)\}}{n_0}$$

- obtain the final variance estimate $\widehat{V}_A / \widehat{ACE}_D^2$

Weak instrumental variable

- $\widehat{CACE}_{\text{Wald}} = \frac{\widehat{ACE}_Y}{\widehat{ACE}_D}$ has poor properties when ACE_D is close to 0
 - $\widehat{CACE}_{\text{Wald}}$ has finite sample bias
 - confidence interval has poor coverage rate

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- How do we deal with a weak instrument (encouragement)?
 - testing: $H_0 : CACE = 0 \iff H'_0 : ACE_Y = 0$
 - confidence interval: Anderson–Rubin type confidence interval based on Fieller (1954)

Anderson–Rubin type confidence interval

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- Inverting hypothesis tests to obtain a confidence interval
- Define the adjusted outcome $A_i(t) = Y_i - tD_i$
 - ACE on $A_i(t)$: $\widehat{\text{ACE}}_A(t)$; variance estimator: $\widehat{V}_A(t)$
 - $\widehat{A}_i(t)$ and $\widehat{V}_A(t)$ are functions of t
 - $\mathbb{E}\{\widehat{\text{ACE}}_A(\text{CACE})\} = 0$
- $\text{pr}\left(\left|\frac{\widehat{\text{ACE}}_A(\text{CACE})}{\sqrt{\widehat{V}_A(\text{CACE})}}\right| \leq 1.96\right) = 0.95$

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- $\text{pr}\left(\left|\frac{\widehat{\text{ACE}}_A(\text{CACE})}{\sqrt{\widehat{V}_A(\text{CACE})}}\right| \leq 1.96\right) = 0.95$
- Solve a quadratic inequality to obtain the confidence interval

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- CACE: effect of the encouragement (participation) on job-search self-efficacy for people who would participate if and only if they are encouraged
- ITT: est. = 0.067, s.e. = 0.050, 95% CI = $[-0.031, 0.166]$
- CACE: est. = 0.109, s.e. = 0.081, 95% CI = $[-0.050, 0.268]$

Covariate adjustment

- We can obtain the adjusted estimator for both the ACEs on D and Y

$$\begin{aligned}\widehat{ACE}_{D,L} &= \left\{ \bar{D}(1) - \widehat{\beta}_{D1}^\top \bar{X}(1) \right\} - \left\{ \bar{D}(0) - \widehat{\beta}_{D0}^\top \bar{X}(0) \right\} \\ \widehat{ACE}_{Y,L} &= \left\{ \bar{Y}(1) - \widehat{\beta}_{Y1}^\top \bar{X}(1) \right\} - \left\{ \bar{Y}(0) - \widehat{\beta}_{Y0}^\top \bar{X}(0) \right\}\end{aligned}$$

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- $\widehat{CACE}_L = \frac{\widehat{ACE}_{Y,L}}{\widehat{ACE}_{D,L}}$

- Asymptotically more efficient than the simple ratio estimator

Variance calculation

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- $\widehat{ACE}_{Y,L} - CACE \cdot \widehat{ACE}_{D,L}$ is the difference in means estimator for the adjusted outcome

$$A_i = \begin{cases} (Y_i - \widehat{\beta}_{Y1}^\top X_i) - CACE \cdot (D_i - \widehat{\beta}_{D1}^\top X_i) & \text{if } Z_i = 1 \\ (Y_i - \widehat{\beta}_{Y0}^\top X_i) - CACE \cdot (D_i - \widehat{\beta}_{D0}^\top X_i) & \text{if } Z_i = 0 \end{cases}$$

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- Variance estimation steps
 - obtain the adjusted outcome
 - obtain the variance estimate based on the adjusted outcome

$$\widehat{V}_{A,L} = \frac{\widehat{\text{var}}\{A_i(1)\}}{n_1} + \frac{\widehat{\text{var}}\{A_i(0)\}}{n_0}$$

- obtain the final variance estimate $\widehat{V}_A / \widehat{ACE}_{D,L}^2$

Evaluation of a job training program

- Do covariate adjustment for the ITT effects
 - center the covariates (without intercept)
 - regress Y on X for both groups to obtain $\hat{\beta}_{Y1}$ and $\hat{\beta}_{Y0}$
 - regress D on X for both groups to obtain $\hat{\beta}_{D1}$ and $\hat{\beta}_{D0}$

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- Calculate the ITT effects: $\widehat{ACE}_{D,L} = 0.617$, $\widehat{ACE}_{Y,L} = 0.059$

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- Calculate the CACE: $\widehat{CACE}_L = \widehat{ACE}_{Y,L} / \widehat{ACE}_{D,L} = 0.096$

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- Calculate the CACE: $\widehat{CACE}_L = \widehat{ACE}_{Y,L} / \widehat{ACE}_{D,L} = 0.096$
- Obtain the adjusted outcome

$$A_i = \begin{cases} (Y_i - \hat{\beta}_{Y1}^\top X_i) - \widehat{CACE} \cdot (D_i - \hat{\beta}_{D1}^\top X_i) & \text{if } Z_i = 1 \\ (Y_i - \hat{\beta}_{Y0}^\top X_i) - \widehat{CACE} \cdot (D_i - \hat{\beta}_{D0}^\top X_i) & \text{if } Z_i = 0 \end{cases}$$

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 - center the covariates (without intercept)
 - regress Y on X for both groups to obtain $\hat{\beta}_{Y1}$ and $\hat{\beta}_{Y0}$
 - regress D on X for both groups to obtain $\hat{\beta}_{D1}$ and $\hat{\beta}_{D0}$
- Calculate the ITT effects: $\widehat{ACE}_{D,L} = 0.617$, $\widehat{ACE}_{Y,L} = 0.059$
- Calculate the CACE: $\widehat{CACE}_L = \widehat{ACE}_{Y,L} / \widehat{ACE}_{D,L} = 0.096$
- Obtain the adjusted outcome

$$A_i = \begin{cases} (Y_i - \hat{\beta}_{Y1}^\top X_i) - \widehat{CACE} \cdot (D_i - \hat{\beta}_{D1}^\top X_i) & \text{if } Z_i = 1 \\ (Y_i - \hat{\beta}_{Y0}^\top X_i) - \widehat{CACE} \cdot (D_i - \hat{\beta}_{D0}^\top X_i) & \text{if } Z_i = 0 \end{cases}$$

- Calculate the variance estimate $\widehat{V}_{A,L} / \widehat{ACE}_{D,L}^2 = 0.006$

Evaluation of a job training program

- Do covariate adjustment for the ITT effects
 - center the covariates (without intercept)
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- Calculate the variance estimate $\widehat{V}_{A,L} / \widehat{ACE}_{D,L}^2 = 0.006$
- CACE: est. = 0.109, s.e. = 0.081, 95% CI = [-0.050, 0.268]
CACE (covs): est. = 0.096, s.e. = 0.080, 95% CI = [-0.061, 0.252]

Effect of veteran status on earnings (Angrist, 1990, *AER*)

- There were five draft lotteries during the Vietnam War period. In each lottery, priority for induction was determined by a Random Sequence Number (RSN) from 1-365 that was assigned to birthdates in the cohort being drafted
- Men were called for induction by RSN up to a ceiling determined by the Defense Department, and only men with lottery numbers below the ceiling could have been drafted
- Draft lottery RSNs were randomly assigned in a televised drawing held a few months before men reaching draft age were to be called
- Draft-eligibility ceilings were announced later in the year, once Defense Department manpower needs were known
- Subsequent selection from the draft-eligible pool was based on a number of criteria: physical examination and a mental aptitude test

IV setup

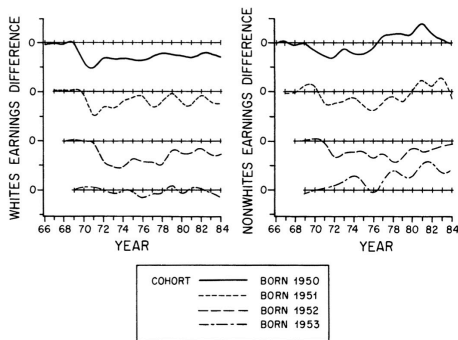
- IV Z_i : draft-eligibility; treatment D_i : veteran status; outcome Y_i : earnings in 1981-1984

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Notes: The figure plots the difference in FICA taxable earnings by draft-eligibility status for the four cohorts born 1950–53. Each tick on the vertical axis represents \$500 real (1978) dollars.

FIGURE 2. THE DIFFERENCE IN EARNINGS BY DRAFT-ELIGIBILITY STATUS

Wald estimates

TABLE 3—WALD ESTIMATES

Cohort	Year	Draft-Eligibility Effects in Current \$			$\hat{p}^e - \hat{p}^n$ (4)	Service Effect in 1978 \$ (5)
		FICA Earnings (1)	Adjusted FICA Earnings (2)	Total W-2 Earnings (3)		
1950	1981	-435.8 (210.5)	-487.8 (237.6)	-589.6 (299.4)	0.159 (0.040)	-2,195.8 (1,069.5)
	1982	-320.2 (235.8)	-396.1 (281.7)	-305.5 (345.4)		-1,678.3 (1,193.6)
	1983	-349.5 (261.6)	-450.1 (302.0)	-512.9 (441.2)		-1,795.6 (1,204.8)
	1984	-484.3 (286.8)	-638.7 (336.5)	-1,143.3 (492.2)		-2,517.7 (1,326.5)
1951	1981	-358.3 (203.6)	-428.7 (224.5)	-71.6 (423.4)	0.136 (0.043)	-2,261.3 (1,184.2)
	1982	-117.3 (229.1)	-278.5 (264.1)	-72.7 (372.1)		-1,386.6 (1,312.1)
	1983	-314.0 (253.2)	-452.2 (289.2)	-896.5 (426.3)		-2,181.8 (1,395.3)
	1984	-398.4 (279.2)	-573.3 (331.1)	-809.1 (380.9)		-2,647.9 (1,529.2)
1952	1981	-342.8 (206.8)	-392.6 (228.6)	-440.5 (265.0)	0.105 (0.050)	-2,502.3 (1,556.7)
	1982	-235.1 (232.3)	-255.2 (264.5)	-514.7 (296.5)		-1,626.5 (1,685.8)
	1983	-437.7 (257.5)	-500.0 (294.7)	-915.7 (395.2)		-3,103.5 (1,829.2)
	1984	-436.0 (281.9)	-560.0 (330.1)	-767.2 (376.0)		-3,323.8 (1,959.3)

Physical activity and weight following car ownership in Beijing, China: quasi-experimental cross sectional study

Michael L Anderson,¹ Fangwen Lu,² Jun Yang³

- In January 2011, to deal with the problem of congestion, Beijing capped the number of new vehicles allowed at 240000 each year and introduced a vehicle permit (license plate) lottery
- After that date, only residents who entered and won the lottery were entitled to a license plate.
- The lottery was drawn monthly, and winners had to purchase a car within six months of winning. By mid-2012 the probability of winning fell below 2% a month

Effect of winning the lottery

- IV: winning the lottery; treatment: buying a car; outcome: weekly transit rides, minute daily walking/bicycling, weight

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Table 3 | Age stratified associations between winning the lottery and transit use, activity, and weight

Dependent variables	Time since winning (95% CI)		
	0.1 years (minimum)	2.6 years (average)	5.1 years (maximum)
Individuals aged ≥40			
Weekly transit rides	-2.18 (-4.13 to -0.24)	-2.1 (-3.35 to -0.85)	-2.02 (-5.16 to 1.12)
Minutes daily walking/bicycling	12.1 (-4.66 to 28.86)	-2.59 (-12.12 to 6.94)	-17.29 (-36.52 to 1.95)
Weight (kg)	1.29 (-5.07 to 7.65)	3.24 (-0.31 to 6.8)	5.2 (-2.59 to 12.99)
Individuals aged ≥50			
Weekly transit rides	-2.88 (-5.57 to -0.19)	-1.9 (-3.61 to -0.18)	-0.91 (-5.45 to 3.63)
Minutes daily walking/bicycling	27.4 (-0.28 to 55.08)	-1.19 (-13.76 to 11.38)	-29.78 (-54.08 to -5.49)
Weight (kg)	-1 (-8.4 to 6.4)	4.67 (0.04 to 9.31)	10.34 (0.49 to 20.19)

Effect of car ownership

- Ratio estimation for the effect of buying a car
- High compliance rate: 0.91 (0.89, 0.94) $\rightsquigarrow$ effect of winning the lottery $\approx$ effect of buying a car

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 - weight is self-reported
 - response rate about 20%
 - target population: people who want to and are able to buy a car

Necessary condition for identification

- Are exclusion restriction and monotonicity necessary?

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 - number of parameters =
$$\begin{cases} 12 & \text{no asm.} \\ 9 & \text{with mon.} \\ 10 & \text{with ex.} \\ 7 & \text{with both} \end{cases}$$
- Without exclusion restriction or monotonicity, CACE is not identifiable

Identification for the joint distribution

- Proportions of principal strata

$$\pi_a = \text{pr}(D = 1 \mid Z = 0), \pi_n = \text{pr}(D = 0 \mid Z = 1), \pi_c = 1 - \pi_a - \pi_n$$

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- $\mathbb{E}\{Y(z) \mid U\} = \mathbb{E}(Y \mid Z = z, U)$ for $U = c$

$$\mathbb{E}(Y \mid z = 1, U = c) = \pi_c^{-1} \{\mathbb{E}(DY \mid Z = 1) - \mathbb{E}(DY \mid Z = 0)\}$$

$$\mathbb{E}(Y \mid z = 0, U = c) = \pi_c^{-1} \{\mathbb{E}((1 - D)Y \mid Z = 0) - \mathbb{E}((1 - D)Y \mid Z = 1)\}$$

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$$\mathbb{E}(Y \mid z = 0, U = c) = \pi_c^{-1} \{\mathbb{E}((1 - D)Y \mid Z = 0) - \mathbb{E}((1 - D)Y \mid Z = 1)\}$$

- Can replace Y with $\mathbf{1}(Y = y)$ to obtain $\text{pr}(Y = y \mid Z = z, U)$

Testable conditions for IV assumptions (Balke & Pearl, 1997)

- Assumptions imply conditions that can be used to falsify the assumptions
- Testable conditions for exclusion restriction and monotonicity

$$\begin{aligned}\text{pr}(Y_i = y, D_i = 1 \mid Z_i = 1) &\geq \text{pr}(Y_i = y, D_i = 1 \mid Z_i = 0) \\ \text{pr}(Y_i = y, D_i = 0 \mid Z_i = 1) &\leq \text{pr}(Y_i = y, D_i = 0 \mid Z_i = 0)\end{aligned}$$

- Statistical test for testable conditions

$$\mathbb{E}(Q \mid Z = 1) - \mathbb{E}(Q \mid Z = 0) \geq 0,$$

where $Q = D \cdot \mathbf{1}(Y = y), (1 - D) \cdot \mathbf{1}(Y = y)$

Examples

- Investigators et al. (2014) assess the effectiveness of the emergency endovascular versus the open surgical repair strategies for patients with a clinical diagnosis of ruptured aortic aneurism
- Instrument $Z = 1$: endovascular strategy; treatment D : treatment received; outcome $Y = 0$: alive
- Statistical tests for testable conditions all pass
- CACE: est. = 0.131; 95% CI: $(-0.036, 0.298)$

Examples

- In Hirano et al. (2000), physicians are randomly selected to receive a letter encouraging them to inoculate patients at risk for flu
- Instrument $Z = 1$: encouragement; treatment D : flu shot; outcome $Y = 0$: flu-shot related hospitalization
- Statistical tests for testable conditions all pass, however,
 $\mathbb{E}\{D(1 - Y) \mid Z = 1\} < \mathbb{E}\{D(1 - Y) \mid Z = 0\}$
- CACE: est. = 0.116; 95% CI: $(-0.061, 0.293)$
- $\mathbb{E}(Y \mid Z = 1, U = c) = 1.004$

CACE in observational studies

- Conditional independence and exclusion restriction may be more plausible after conditioning on covariates $\mathbf{X}_i$
 - $Z_i \perp\!\!\!\perp \{Y_i(1), Y_i(0), D_i(1), D_i(0)\} \mid \mathbf{X}_i$
 - $Y_i(1) = Y_i(0)$ for $U = a, n$

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- Within subgroup defined by $\mathbf{X}$, estimate
$$\text{CACE}(\mathbf{x}) = \mathbb{E}\{Y_i(1) - Y_i(0) \mid D_i(1) = 1, D_i(0) = 0, \mathbf{X}_i = \mathbf{x}\}$$

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- Estimate $\text{CACE}(\mathbf{x})$ for each $\mathbf{x}$ and then average over $\mathbf{X}$

$$\text{CACE} = \mathbb{E}\{\text{CACE}(\mathbf{X}_i) \mid D_i(1) = 1, D_i(0) = 0\},$$

which requires the estimation of $\text{pr}(\mathbf{X}_i \mid D_i(1) = 1, D_i(0) = 0)$

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which requires the estimation of $\text{pr}(\mathbf{X}_i \mid D_i(1) = 1, D_i(0) = 0)$

- For continuous $\mathbf{X}_i$, we need to model $\text{CACE}(\mathbf{x})$, which is hard to estimate because the model is for compliers only

Weighting method for CACE

Theorem (Abadie Kappa)

Suppose that the assumptions for CACE hold conditional on $\mathbf{X}_i$. Define

$$\begin{aligned}\kappa_{0i} &= (1 - D) \frac{(1 - Z) - \text{pr}(Z = 0 \mid \mathbf{X}_i)}{\text{pr}(Z = 0 \mid \mathbf{X}_i) \text{pr}(Z = 1 \mid \mathbf{X}_i)}, \\ \kappa_{1i} &= D \frac{Z - \text{pr}(Z = 1 \mid \mathbf{X}_i)}{\text{pr}(Z = 0 \mid \mathbf{X}_i) \text{pr}(Z = 1 \mid \mathbf{X}_i)}.\end{aligned}$$

Then

$$\begin{aligned}\mathbb{E}\{g(Y_i(0), \mathbf{X}_i) \mid U_i = c\} &= \frac{1}{\text{pr}\{U_i = c\}} \mathbb{E}\{\kappa_{0i} g(Y, X)\} \\ \mathbb{E}\{g(Y_i(1), \mathbf{X}_i) \mid U_i = c\} &= \frac{1}{\text{pr}\{U_i = c\}} \mathbb{E}\{\kappa_{1i} g(Y, X)\}\end{aligned}$$

Abadie's procedure for estimating CACE

- Two-step procedure
 - step 1: fit a model for $\text{pr}(Z_i = 1 \mid \mathbf{X}_i)$, and calculate κ_{0i} and κ_{1i} for each unit
 - step 2: apply Abadie's formula

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- Abadie's procedure simplifies to Wald estimator without covariates

Model-assisted weighting method

- We may want to assume some parametric models for $Y(z)$, e.g.
 $\mathbb{E}\{Y_i(z) \mid \mathbf{X}_i, D_i(1) = 1, D_i(0) = 0\} = \beta_{z0} + \beta_{z,X}\mathbf{X}_i$
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- Least squares for compliers:

$$\begin{aligned} & \underset{\beta}{\operatorname{argmin}} \mathbb{E} \left[\{Y_i(z) - (\beta_{z0} + \beta_{z,X}\mathbf{X}_i)\}^2 \mid D_i(1) = 1, D_i(0) = 0 \right] \\ & \equiv \underset{\beta}{\operatorname{argmin}} \mathbb{E} \left[\kappa_{zi} \{Y_i - (\beta_{z0} + \beta_{z,X}\mathbf{X}_i)\}^2 \right] \end{aligned}$$

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- Abadie's procedure
 - step 1: fit a model for $\operatorname{pr}(Z_i = 1 \mid \mathbf{X}_i)$, and calculate κ 's for each unit
 - step 2: weighted least squares estimation

Summary

- ITT vs. CACE $\rightsquigarrow$ additional assumptions are required
 - ① randomization of instrument
 - ② monotonicity
 - ③ exclusion restriction
- Problems of external validity:
 - compliers vs. non-compliers
 - compliers as latent group defined by an instrument
- Exclusion restriction and monotonicity imply testable conditions

Suggested readings

- Non-compliance
 - Angrist, Imbens & Rubin, “Identification of Causal Effects Using Instrumental Variables”
 - IMBENS AND RUBIN, Chapters 23 and 24
 - ANGRIST AND PISCHKE, Chapter 4
- Weighting method for CACE
 - Abadie (2003). “Semi-parametric instrumental variable estimation of treatment response models”